mutually exclusive. If the spin is in one of these states, it *cannot* be (has zero probability to be) in the other one. This idea applies to all quantum systems, not just spin.

But don't mistake the orthogonality of state-vectors for orthogonal directions in space. In fact, the directions *up* and *down* are not orthogonal directions in space, even though their associated state-vectors are orthogonal in state space.

The second important point is that for the total probability to come out equal to unity, we must have

$$\alpha_u^* \alpha_u + \alpha_d^* \alpha_d = 1. \qquad (2.4)$$

This is equivalent to saying that the vector |*A*> is normalized to a unit vector:

$$\langle A|A\rangle = 1.$$

This is a very general principle of quantum mechanics that extends to all quantum systems: the state of a system is represented by a unit (normalized) vector in a vector space of states. Moreover, the squared magnitudes of the components of the state-vector, along particular basis vectors, represent probabilities for various experimental outcomes.

2.3 ALONG THE X AXIS

We said before that we can represent any spin state as a linear combination of the basis vectors |*u*> and |*d*>. Let's try doing this now for the vectors |*r*> and |*l*>, which represent spins prepared along the *x* axis. We'll start with |*r*>. As you recall from Lecture 1, if $\mathcal{A}$ initially prepares |*r*>, and is then rotated to measure σ_z, there will be equal probabilities for *up* and *down*. Thus, $\alpha_u^* \alpha_u$ and $\alpha_d^* \alpha_d$ must both be equal to ½. A simple vector that satisfies this rule is